

Lecture 8: OSI Model — In Depth

The Data Link Layer handles MAC addressing, ARP, framing, and flow/access control — and the Network Layer builds on top of it with four core responsibilities of its own.

MAC Address — In Depth

MAC Address = Media Access Control Address, also known as the **Physical Address**.

- ♦ It is **unique** to each device and **does not change** under normal circumstances (it can technically be changed/spoofed, but this is considered **illegal**).
- ♦ Every device — laptop, phone, printer, etc. — has a MAC address, and the number of MAC addresses a device has depends on its NIC (the number of ways it can connect to the internet).

Format of MAC Address

48-bit number, represented in Hexadecimal form

Minimum: 00:00:00:00:00:00

Maximum: FF:FF:FF:FF:FF:FF

Each hex digit = 4 bits

6 groups × 8 bits per group = 48 bits total

How MAC Addresses Are Distributed

24 Bits

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- ♦ The **first 24 bits** are assigned to a specific **organization** (manufacturer).
- ♦ The **remaining 24 bits** let that organization create up to **2^{24} unique devices**.

Examples:

ORGANIZATION	ASSIGNED PREFIX
Cisco	CC : 46 : D6
Google	3C : 5A : B4
Hewlett Packard	3C : D9 : 2B

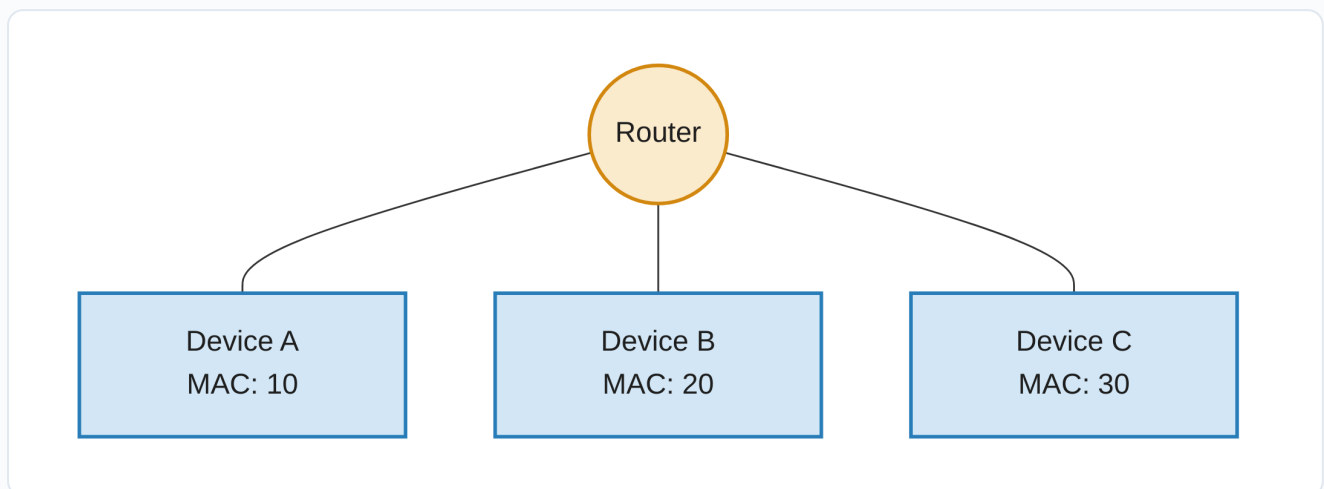
Each organization can generate **2^{24} devices** starting from its assigned prefix — a MAC address is therefore a **12-digit hexadecimal number**, which is fundamentally a **48-digit binary number**.

Notations of MAC Address

MAC addresses can be written in three equivalent formats:

1. Hyphen-Hexadecimal: 00-0a-83-b1-c0-8e
2. Colon-Hexadecimal: 00:0a:83:b1:c0:8e
3. Period-Separated Hexadecimal: 000.a83.b1c.08e

How Are Other Devices' MAC Addresses Known Within a LAN?



- ♦ **Device A** can transfer/send data using the **MAC address** of any device in that particular network.
- ♦ The **router assigns Private IP Addresses**, and those private IPs are known to the devices — but their **MAC addresses** still need to be discovered separately.

ARP — Address Resolution Protocol

ARP (Address Resolution Protocol) finds the **MAC address** of a device when only its **IP address** is known.

Scenario: Device A wants to send data to Device B, but only knows B's **IP address**, not its MAC address.

1. **A broadcasts** a message using B's IP address, requesting: "Whoever has this IP, please respond with your MAC address."
2. **B recognizes** its own IP in the request and **responds with its MAC address**.
3. Now A has B's MAC address and can **send data directly**.

How Does an ISP Trace Illegal Activity to a Specific Device?

A device's IP address can change over time — especially with a **dynamic IP address** — so the IP address alone isn't enough to permanently identify a device. To solve this, the ISP also keeps track of each device's **MAC address**, since a device connects directly to the ISP's network hardware and its MAC address stays visible at that point.

- ✦ The ISP maintains a log of which IP address was assigned to which MAC address, at what time.
- ✦ If illegal activity is traced back to a specific IP address, the ISP can check this log to find out exactly which device (identified by its MAC address) was using that IP address at that particular time.

Error Detection & Correction

A responsibility of the Data Link Layer — detecting when a **bit error** occurs during transmission (e.g. a 0 flips to a 1), and either **correcting the error** or requesting a **resend**.

Bit sent: 0 – 0 – 0
Bit received: 0 – 1 – 0 ← error occurred

DLL response: correct the error, or ask sender to resend the data.

Framing

Framing defines the **structure** in which data actually travels across the Data Link Layer.



How does the receiver know when data is about to arrive?

- ✦ Normally, the signal on the line is **idle** (represented as all 000000).
- ✦ When the sender is about to transmit a message, it sends a distinctive **alert signal**, e.g. 1010101011.

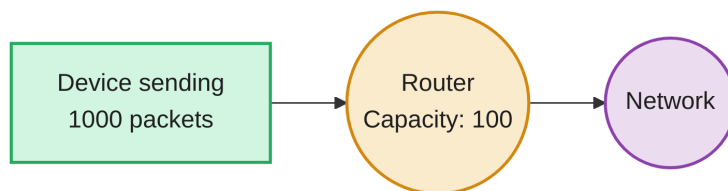
- ♦ As soon as the receiver detects this **pattern** (specifically, recognizing 11 at the end), it switches into an **"alert mode"** and prepares to read the **actual incoming data**.

[A] --- 1010101...11 ---> [B] (B becomes alert and ready to receive)

This pattern-based signaling is what tells the receiver: "real data is coming right after this."

Flow Control

*Ensures that data is sent to a device **at a rate it can actually handle**, preventing overload.*

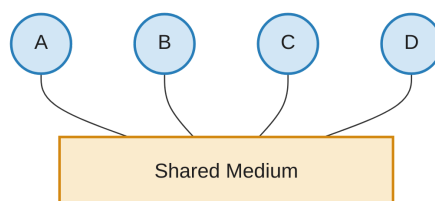


- ♦ If a router's capacity is only **100 packets**, but a device sends **1000 packets** at once, the router **cannot handle the excess**, leading to **data loss**.
- ♦ **Solution:** don't send all packets at once — send the **desired amount of data in a controlled sequence**.

Flow Control = sending a controlled amount of data to a device, in proper sequence, matching the receiver's capacity.

Access Control

*Manages **which device gets to transmit at any given moment** when multiple devices share the same medium.*



- ♦ If multiple devices try to send signals over the same medium **simultaneously**, their signals can **collide**, potentially **corrupting the data**.
- ♦ **Solution:** allow **only one device to transfer data at a time**.

Example order: D → C, then B → A (one transmission completes before the next starts)

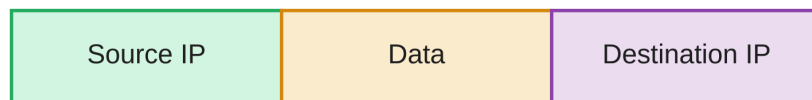
*This controlling of device access is handled by the **Data Link Layer** and is called **Accessing** — it improves the overall **efficiency of the shared cables/medium**.*

3. Network Layer

The Network Layer has **4 core responsibilities**:

1) Assigning Logical Address

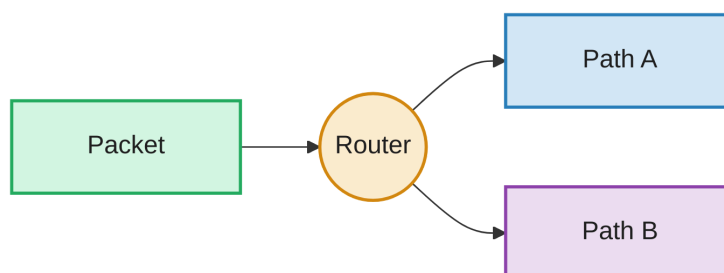
Logical Address = IP Address — this is the layer that gives a device its IP Address.



This is what connects the sender and receiver across the full network — unlike the Data Link Layer, which only handles a single hop at a time.

2) Routing

Routing = deciding the **route/path** a packet should take to reach its destination.



A router operates using 3 layers — Network, Data Link, and Physical — and the routing decision (which direction to send the packet) is made at the **Network Layer**.

3) Fragmentation & Re-Assembly

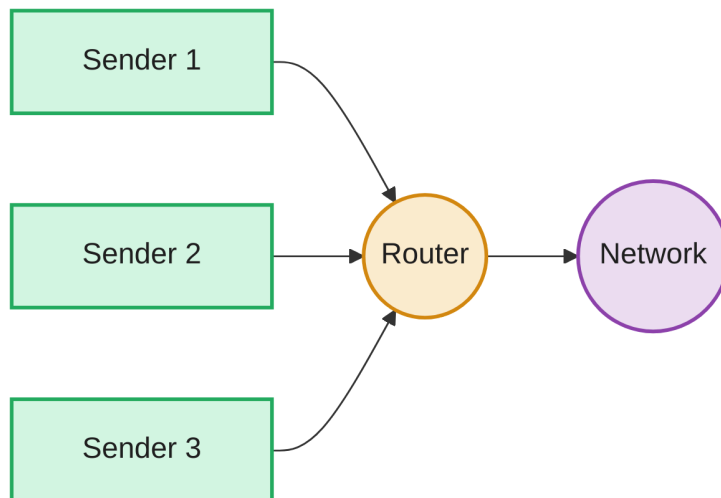
Data Link Layer unit: Frame
Network Layer unit: Packet

Scenario: A packet of data (e.g. 5 MB) needs to travel across a network segment that can only handle a smaller capacity (e.g. 2.5 MB at a time).



- ✦ The original packet is **divided into smaller fragments** to fit within the network's capacity.
- ✦ **After reaching the destination, these fragments are re-assembled** back into the original data.

4) Congestion Control



- ✦ If **all senders transmit data at once**, excessive **traffic** is created, leading to:
 - ⚠ **Delay**
 - ⚠ **Collision**
 - ⚠ **Data loss**

- ♦ **Solution:** there needs to be a **systematic way (protocols)** that governs how and when data is sent, to avoid overwhelming the network.

Key Terms:

TERM	MEANING
Bandwidth	The maximum speed at which data can theoretically be transmitted over a network
Throughput	The actual speed at which data is currently being transferred (often lower than bandwidth due to real-world conditions)

Summary

- ♦ A MAC Address (Media Access Control Address / Physical Address) is a unique, generally unchanging 48-bit hexadecimal identifier assigned to each device's network interface, with the first 24 bits identifying the manufacturer and the remaining 24 bits identifying the specific device.
- ♦ MAC addresses can be written in hyphen-hexadecimal, colon-hexadecimal, or period-separated hexadecimal notation — all representing the same underlying value.
- ♦ ARP (Address Resolution Protocol) resolves a device's MAC address from its known IP address by broadcasting a request across the local network, to which the target device responds with its MAC address.
- ♦ The Data Link Layer also handles Error Detection & Correction (catching and fixing/resending corrupted bits), Framing (structuring data with a recognizable alert signal so the receiver knows when real data is arriving), Flow Control (sending data at a rate the receiver can handle), and Access Control (ensuring only one device transmits over a shared medium at a time to avoid collisions).
- ♦ The Network Layer has four core jobs: assigning the Logical (IP) Address to connect sender and receiver, Routing (deciding a packet's path, done at the router), Fragmentation & Re-Assembly (splitting large packets to fit network capacity and reconstructing them at the destination), and Congestion Control (managing traffic to avoid delay, collisions, and data loss).
- ♦ Bandwidth is the maximum possible data transmission speed over a network, while Throughput is the actual speed being achieved in practice.